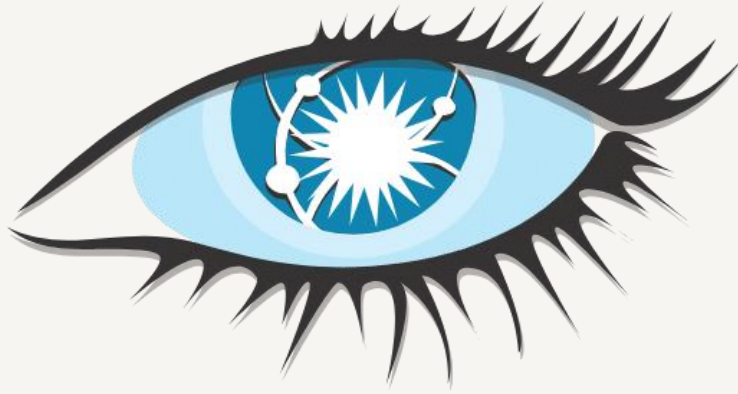




Set up Hybrid RAG retrieval on Apache Cassandra

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SEPTEMBER 17TH, 2026

Mariah McLaughlin

- Developer Advocate: NetApp Instaclustr
- Co-Host NYC Open Data User Group
- Co-Host Bi-Monthly Planet Cassandra Virtual Meetups
- Previously: Software engineer Nextmv (FICO) and Vultr
- Unintentionally works for companies opposed to vowels
-  **mamclaughlin**

Key Takeaways

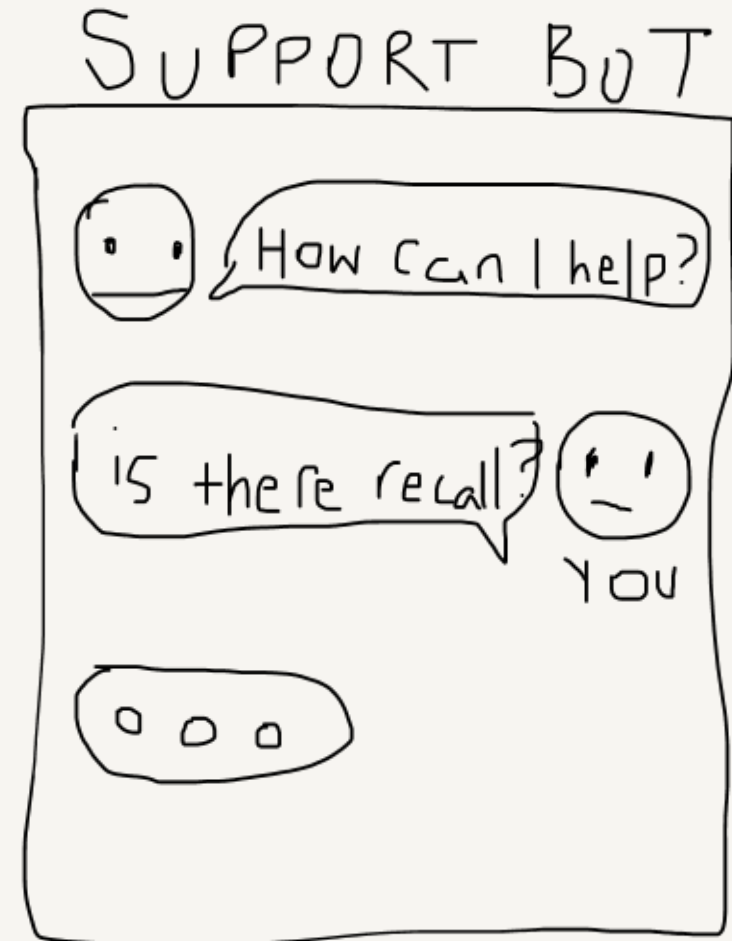
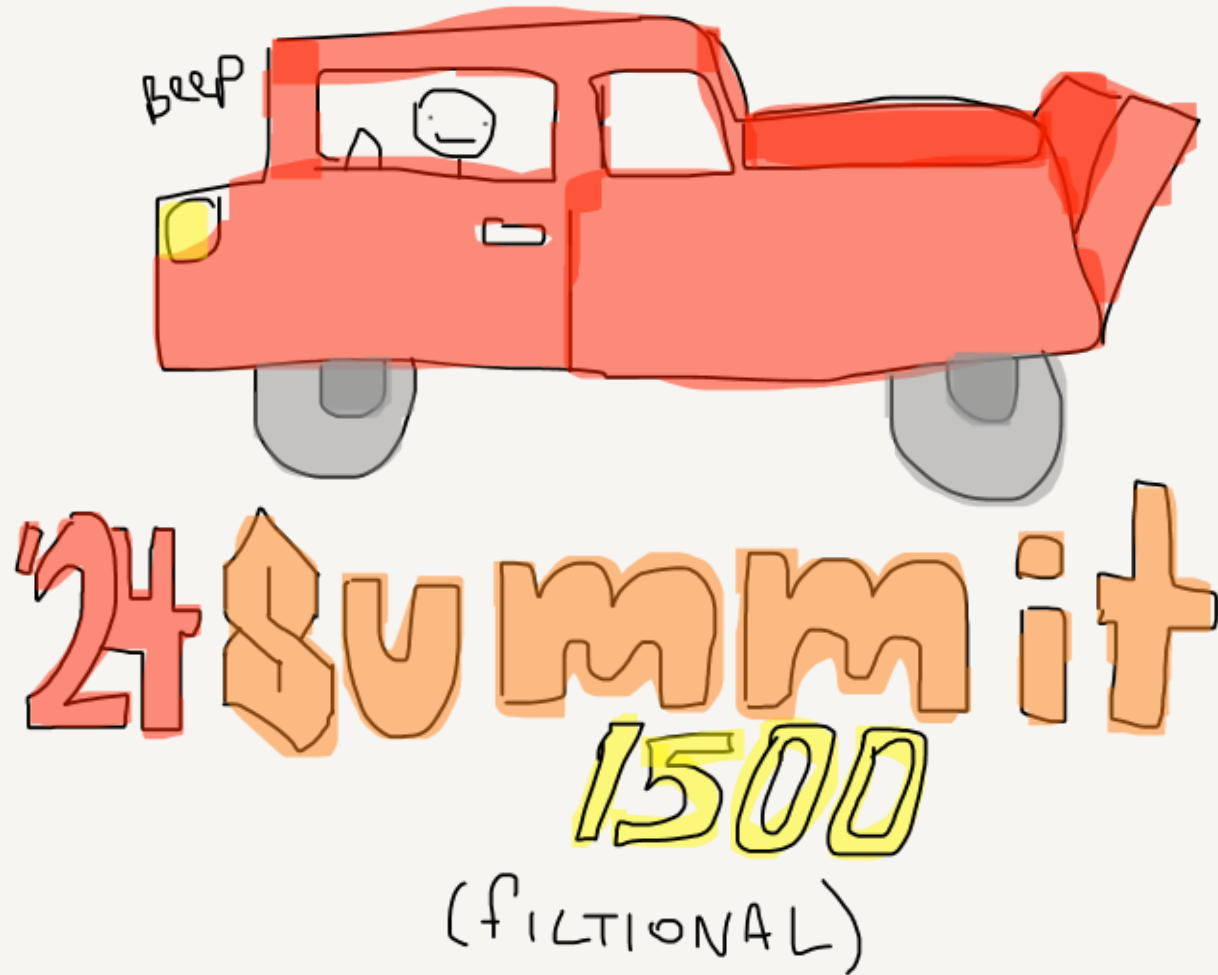
How to set up vector and hybrid-like retrieval on Apache Cassandra

AI search uses vectors for information retrieval, but how we use vectors determines quality of our answers

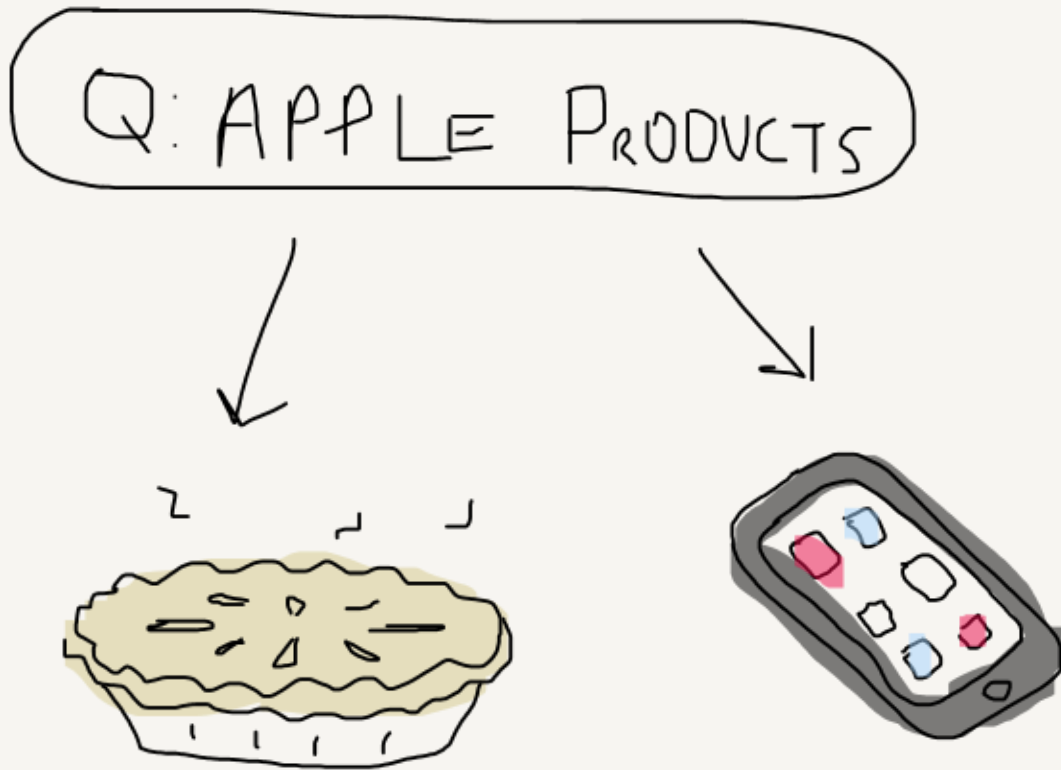
Understand vector limitations, and strategies to improve correctness before you reach the LLM

Is the best vector database the one you already use? Sometimes! Let's find out.

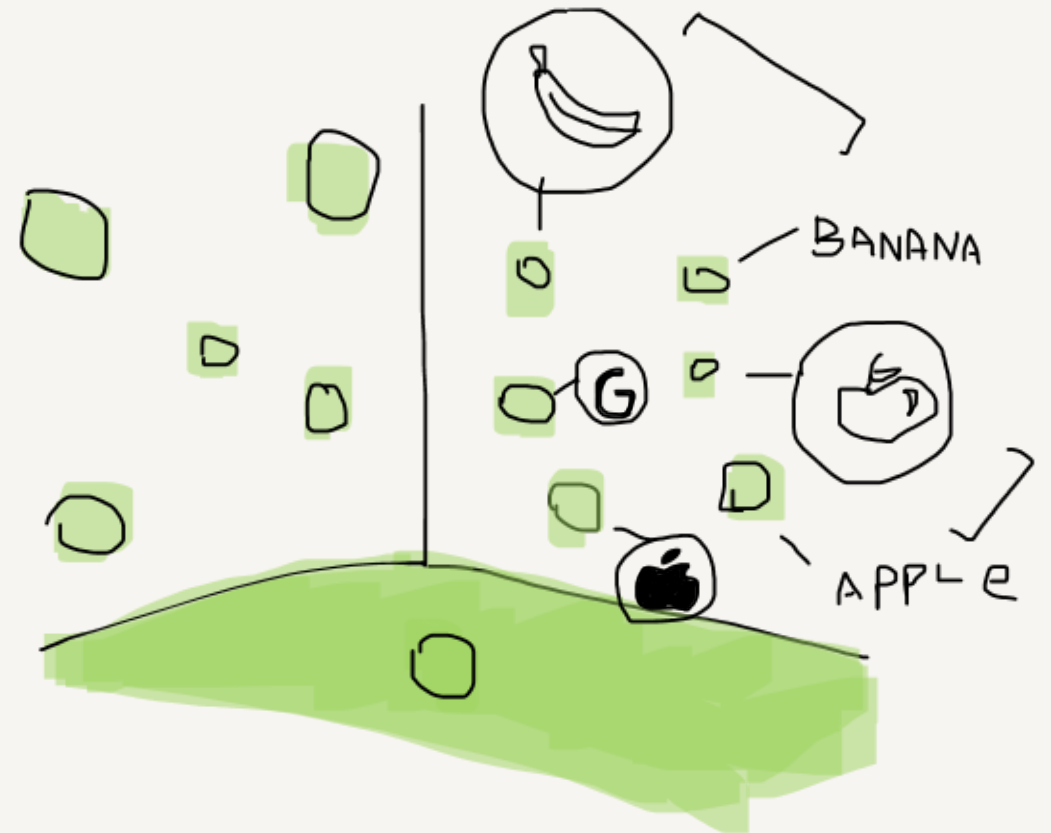
Set the scene



How we search: Before & Now



Lexical, keyword search – lacks context

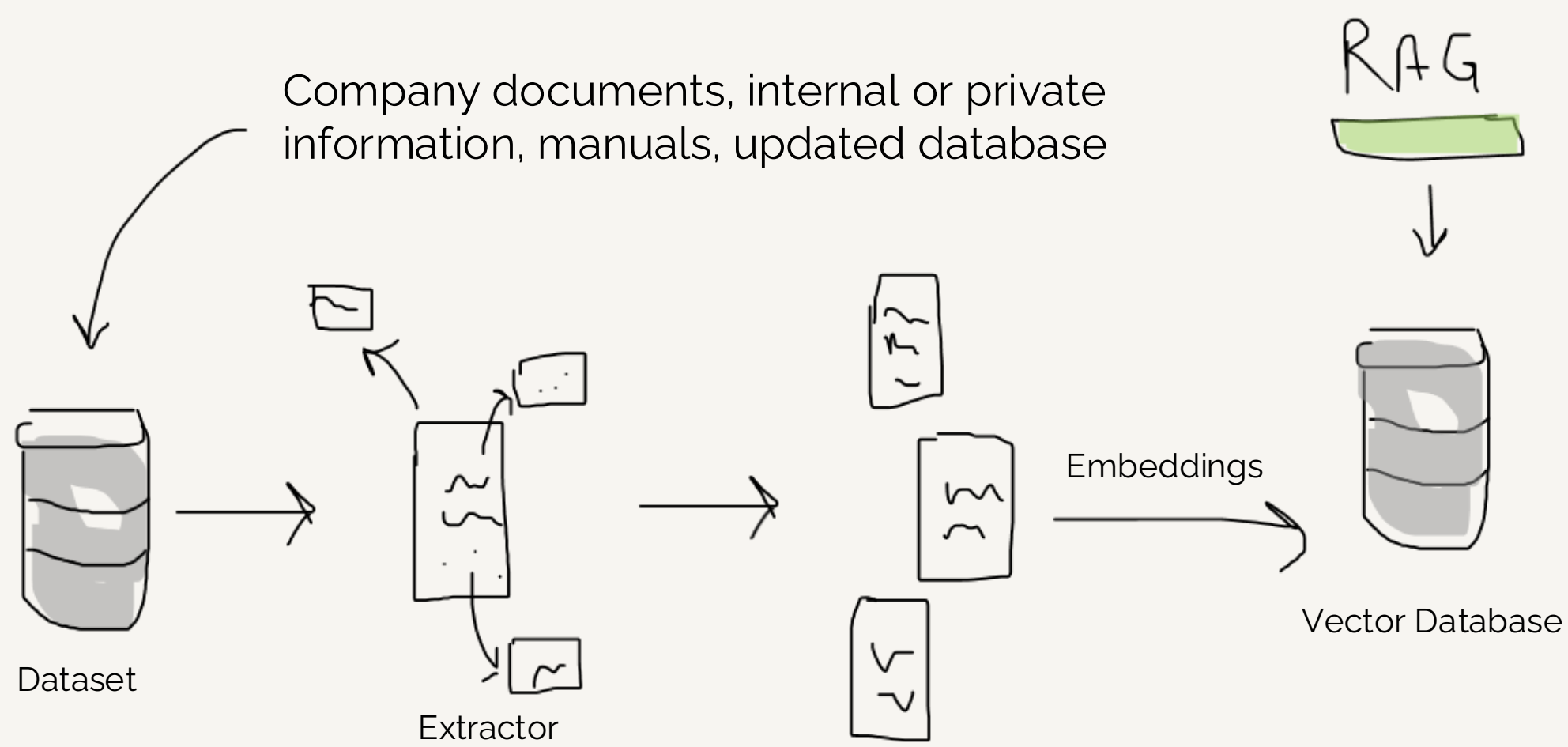


Semantic meaning, building relationships

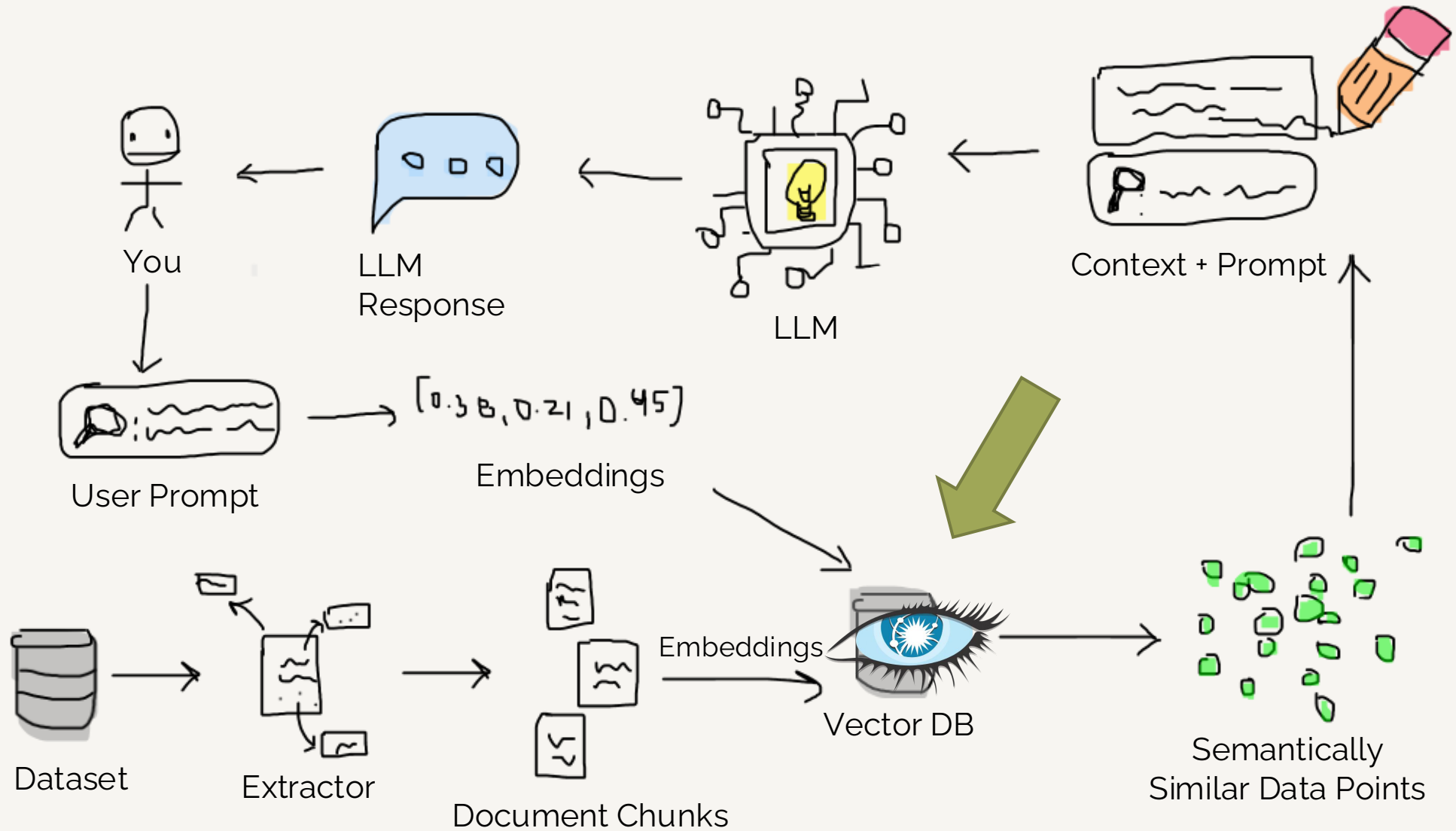
Vector search

- Basis of semantic similarity - it captures relations between concepts: cats vs felines
- Context aware – apple the fruit vs Apple the company
- Removes the need for developers to build large synonym or cross-language dictionaries
- Enables **Retrieval Augmented Generation (RAG)**

Retrieval Augmented Generation (RAG)

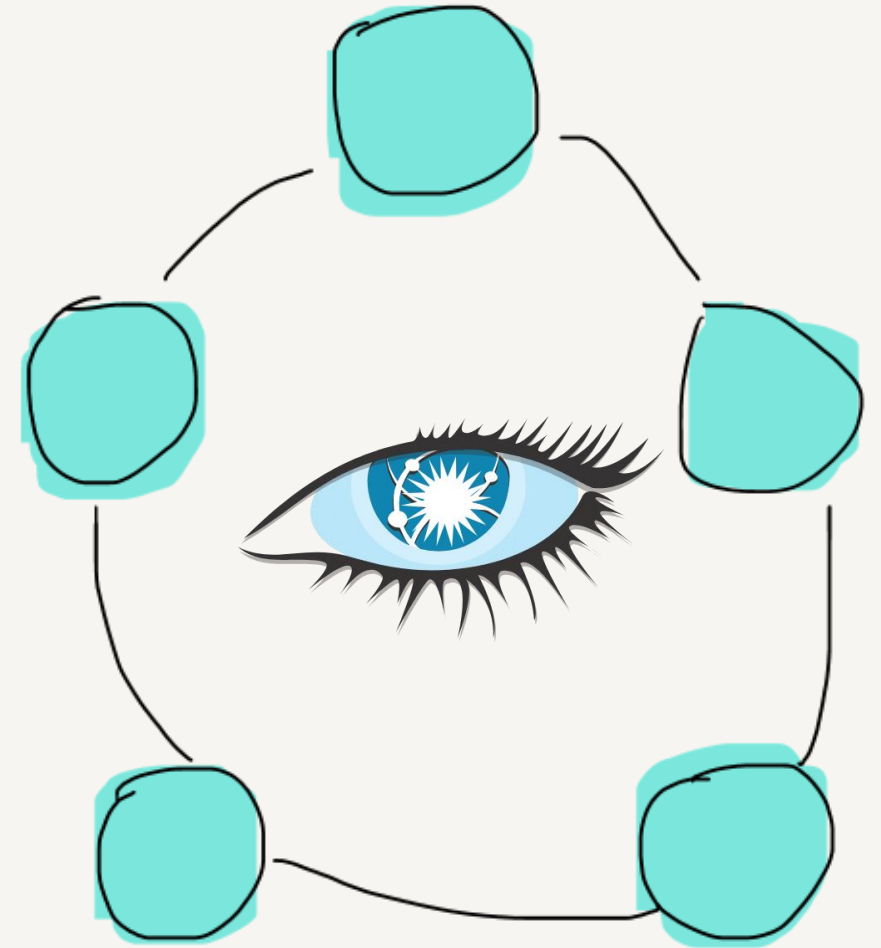


Where retrieval sits in a RAG pipeline



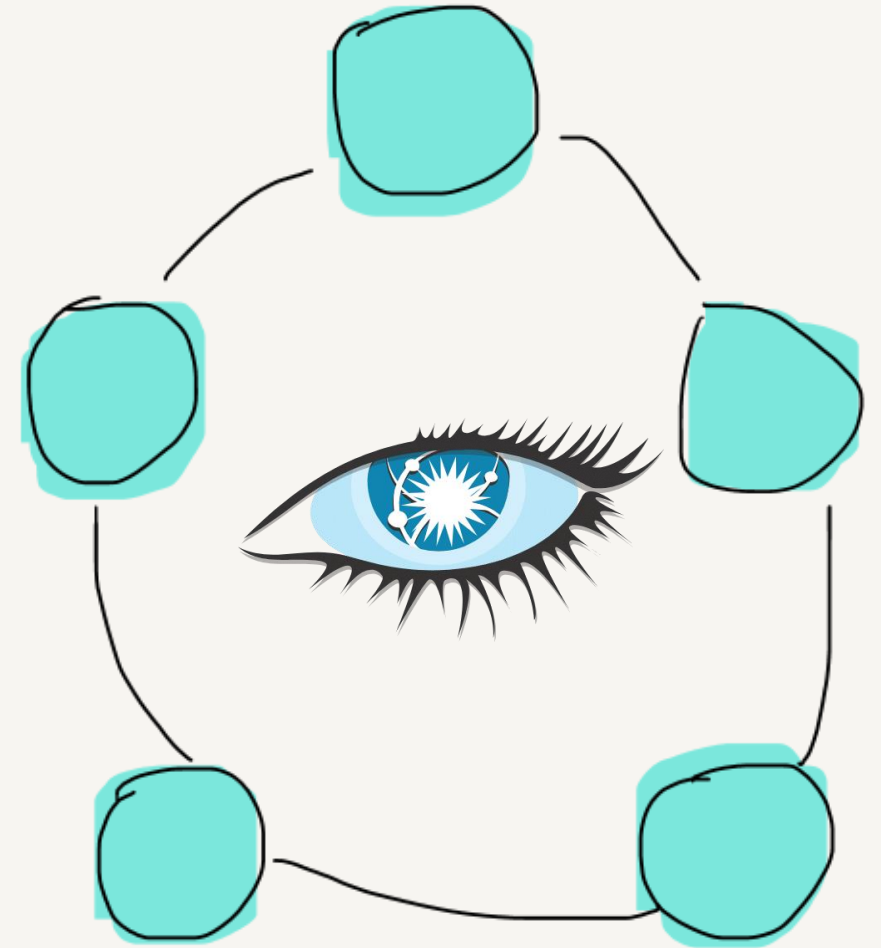
Cassandra in a nutshell

- Distributed database
- Handles heavy writes
- Masterless: every node is a peer
- Limitations: no Joins, Query-led design
- CQL (Cassandra Query Language)
- Users: Netflix, Apple, Bloomberg, Spotify, Walmart, and more!



Cassandra 5 & AI

- Cassandra 5 introduced **Storage-Attached Indexing** and the **vector** data type (2024)
- **vector<float, N>**: column type holding an embedding of N dimensions, sitting on the row it describes rather than in another system.
- **SAI (Storage-Attached Indexing)** attaches a searchable index to the table as rows are written.
- All together - Cassandra now becomes AI search enabled



CQL Setup

```
CREATE TABLE docs (  
  id text PRIMARY KEY,  
  title text,  
  body text,  
  category text,  
  model text,  
  model_year int,  
  embedding vector<float, 384>  
);
```

```
CREATE INDEX ON docs (embedding) USING 'sai'  
  WITH OPTIONS = {'similarity_function': 'COSINE'};
```

```
-----  
SELECT id, title, category, model, model_year,  
       similarity_cosine(embedding, ?) AS score  
FROM docs  
ORDER BY embedding ANN OF ?  
LIMIT 5;
```

That's all you need to get started

Demo

```
(.venv) mm11621@mm11621-mac-0 cassandra-vector-demo % python src/setup_schema.py
Waiting for Cassandra...
Applying vector schema and ANN index...
Done. Keyspace: vector_demo Table: docs
(.venv) mm11621@mm11621-mac-0 cassandra-vector-demo % python src/seed.py
Loaded 21 docs from /Users/mm11621/Documents/code-samples/Cassandra/cassandra-vector-demo/data/corpus.json
Embedding with sentence-transformers (first run downloads the model)...
|
```

Everyday paraphrase finds the right recall

"Is there a recall on the tailgate?"

```
(.venv) mm11621@mm11621-mac-0 cassandra-vector-demo % python src/demo.py "Is there a recall on the tailgate?"  
[]
```

Everyday paraphrase finds the right recall

"Is there a recall on the tailgate?"

```
=====
QUERY: Is there a recall on the tailgate?
Loading weights: 100%|██████████████████████████████████████████| 103/103 [00:00<00:00, 18680.04it/s]
1. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch score=0.7767
2. [recalls] recall-23v-011 [1500 2023] Recall 23V-011 instrument cluster score=0.7032
3. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera score=0.6935
4. [recalls] recall-23v-145 [2500 2023] Recall 23V-145 brake booster score=0.6934
5. [recalls] recall-22v-078 [2500 2022] Recall 22V-078 trailer brake module score=0.6808
```



Sometimes, an exact recall ID lands nowhere near the top

What is recall 24V-330?

```
(.venv) mm11621@mm11621-mac-0 cassandra-vector-demo % python src/demo.py "Is there a recall on the tailgate?"  
"
```

Sometimes, an exact recall ID lands nowhere near the top

What is recall 24V-330?

```
=====
QUERY: What is recall 24V-330?
 1. [recalls] recall-23v-011 [1500 2023] Recall 23V-011 instrument cluster score=0.7871
 2. [recalls] recall-24v-207 [1500 2024] Recall 24V-207 fuel pump relay score=0.7701
 3. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera score=0.7623
 4. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch score=0.7560
 5. [recalls] recall-24v-155 [2500 2024] Recall 24V-155 glow plug controller score=0.7514
```



A question retrieval cannot answer what's being asked

How much can the Summit 1500 tow?

```
(.venv) mm11621@mm11621-mac-0 cassandra-vector-demo % python src/demo.py "How much can the Summit 1500 tow?"
```

Let's take a look at what the data says...

A question retrieval cannot answer what's being asked

How much can the Summit 1500 tow?

=====

QUERY: How much can the Summit 1500 tow?

1. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year score=0.8990
2. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year score=0.8853
3. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity score=0.8791
4. [specs] summit-1500-2024-payload [1500 2024] 2024 Summit 1500 payload capacity score=0.8276
5. [towing] hitch-classes Hitch receiver classes and weight distribution score=0.7550



Let's take a look at what the data says...

How much can the Summit 1500 tow?

```
{
  "id": "summit-1500-2023-tow",
  "title": "Tow ratings by configuration, 2023 model year",
  "category": "towing",
  "model": "1500",
  "model_year": 2023,
  "body": "The Summit 1500 tows up to 9,100 lb with the towing package, which covers most travel trailers and a two-car open trailer. Figures on this page describe the 2023 model year."
},
```

Same model,
different year,
different weights

```
{
  "id": "summit-1500-2024-tow",
  "title": "Tow ratings by configuration, 2024 model year",
  "category": "towing",
  "model": "1500",
  "model_year": 2024,
  "body": "Ratings for the 2024 Summit 1500 vary by driveline. With the 3.5L V6, 3.73 axle, and Max Tow Package, the published maximum is 11,300 lb against a GCWR of 17,000 lb. Lower axle ratios and the base cooling package derate this figure."
},
```

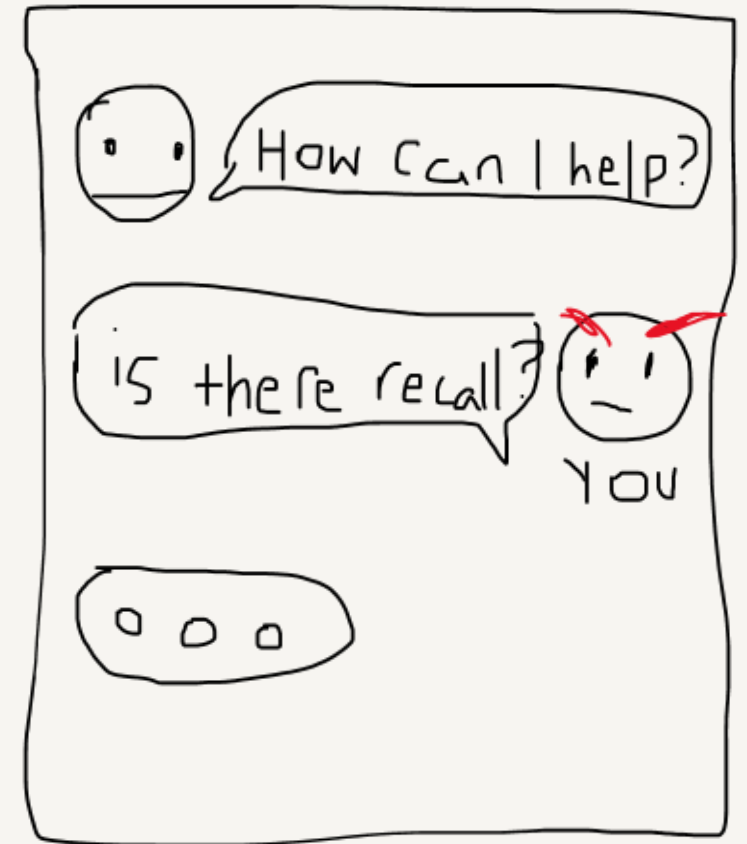


Semantic search != Answer

Vector search is enough when the question is about meaning and an incorrect response is recoverable

Not enough when an identifier, a model year, or a customer record has to be exact

Those need a lexical path or a WHERE clause next to ANN, using additional SAI indexes on this same table



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arXiv:2408.04948v1 [cs.CL] 09 Aug 2024

Abstract.

1 Introduction

2 Methodology

3 Data Description

4 Implementation

Details

5 Results

6 Conclusion and
Future Directions

7 Acknowledgement

References

HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction

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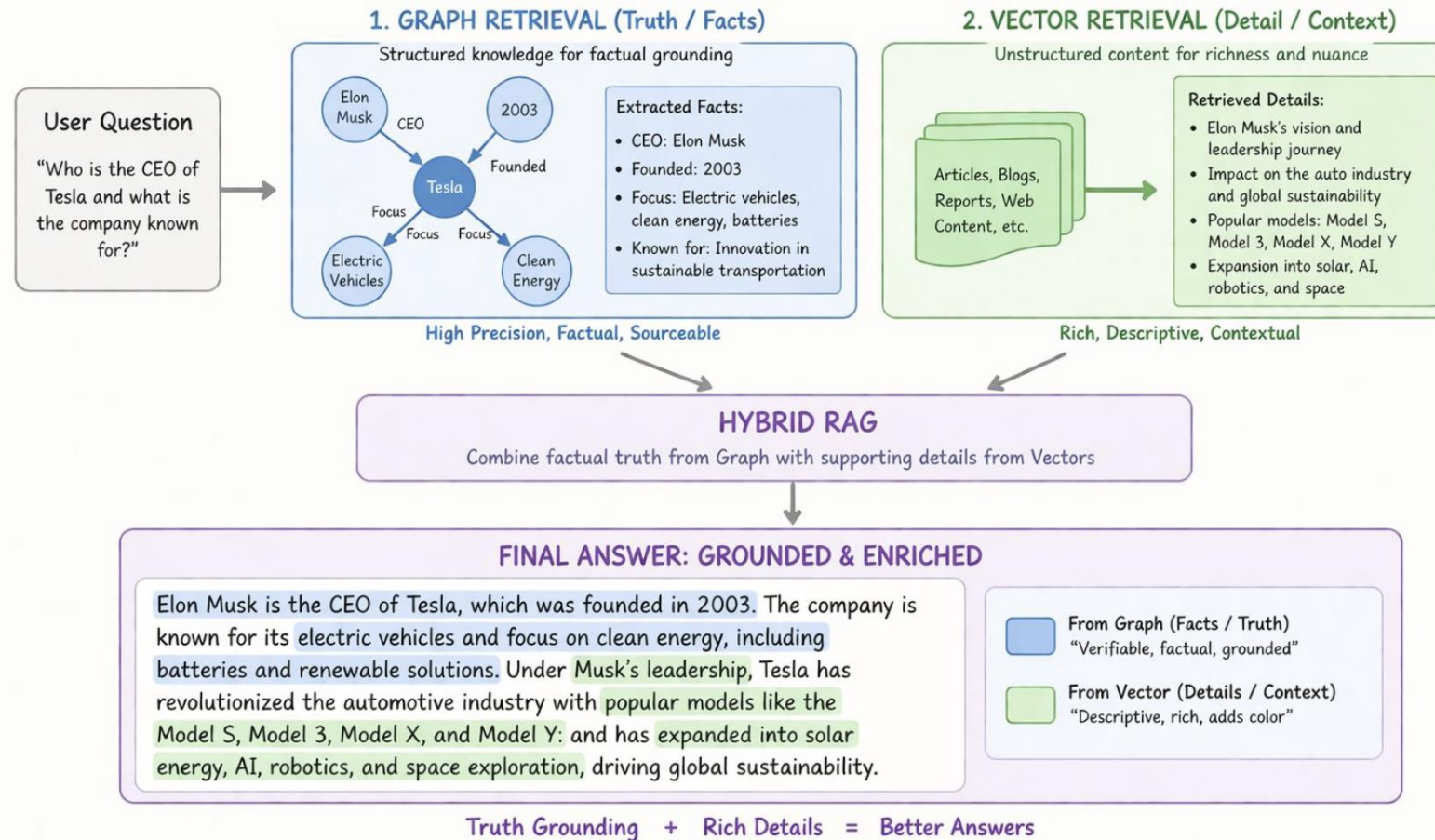
Abstract.

Extraction and interpretation of intricate information from unstructured text data arising in financial applications, such as earnings call transcripts, present substantial challenges to large language models (LLMs) even using the current best practices to use Retrieval Augmented Generation (RAG) (referred to as VectorRAG technique which utilize vector databases for information retrieval) due to challenges such as domain specific terminology and complex formats of the documents. We introduce a novel approach based on a combination, called *HybridRAG*, of the

[HTTPS://ARXIV.ORG/HTML/2408.04948V1](https://arxiv.org/html/2408.04948v1)

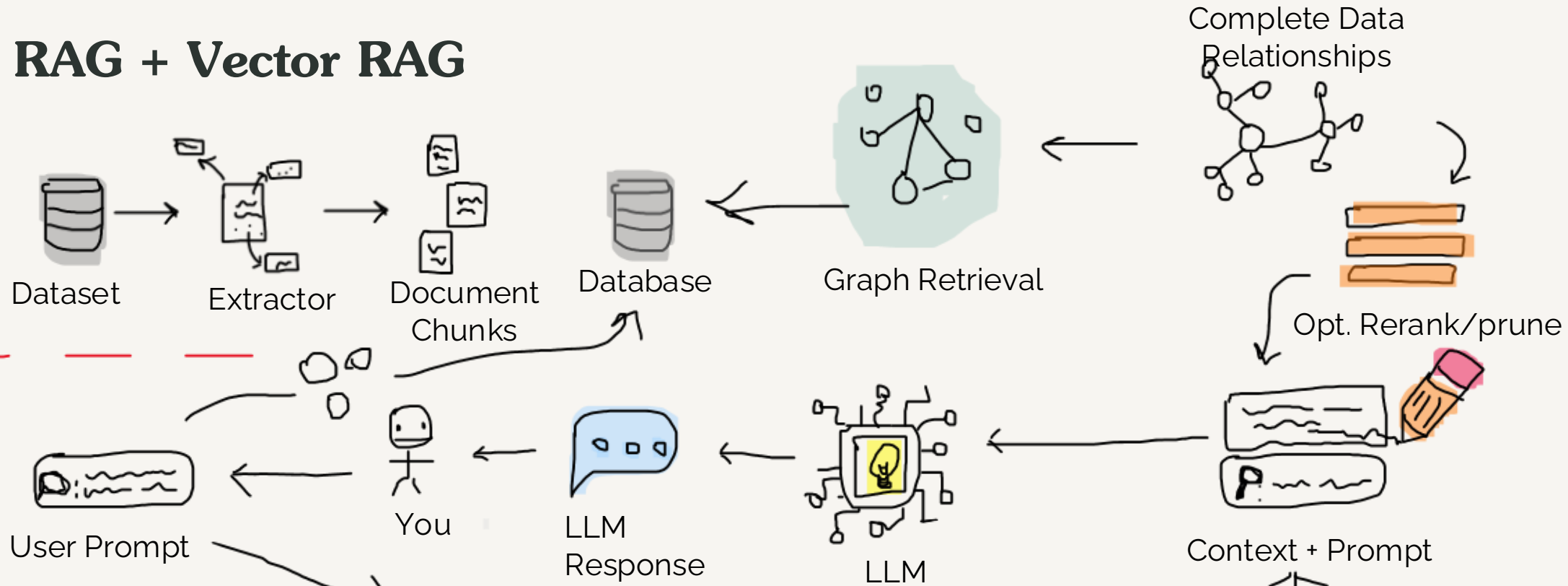
Hybrid RAG: Truth Grounding + Adding Detail

Use Graph Retrieval for factual accuracy and Vector Embeddings for richness and context.

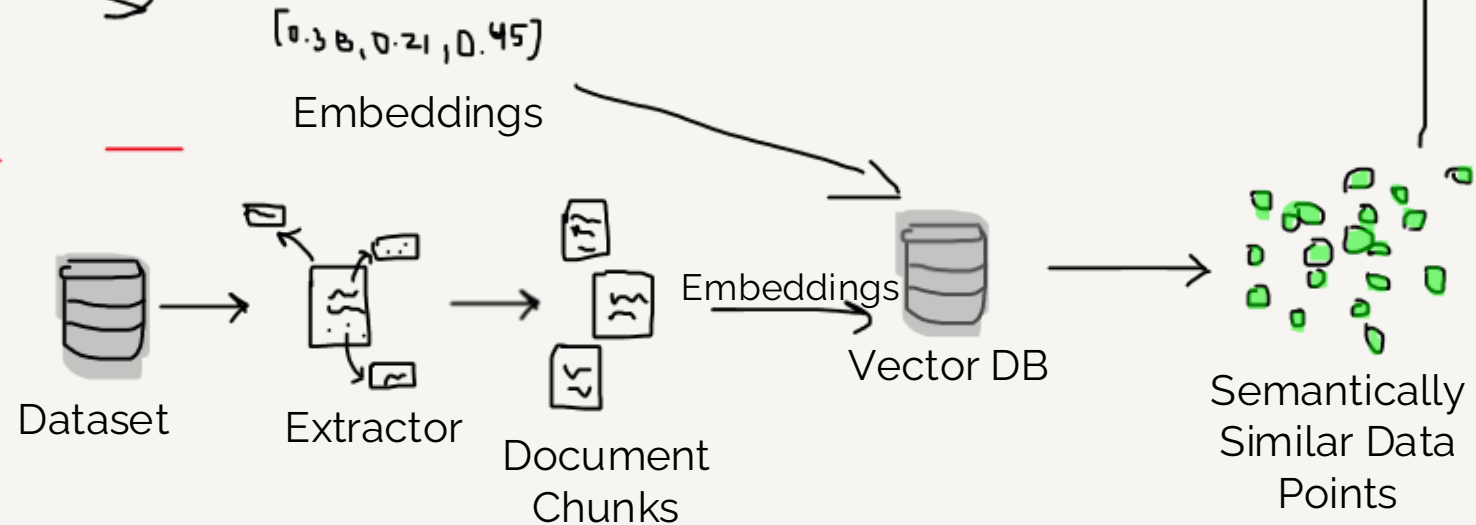


Graph RAG + Vector RAG

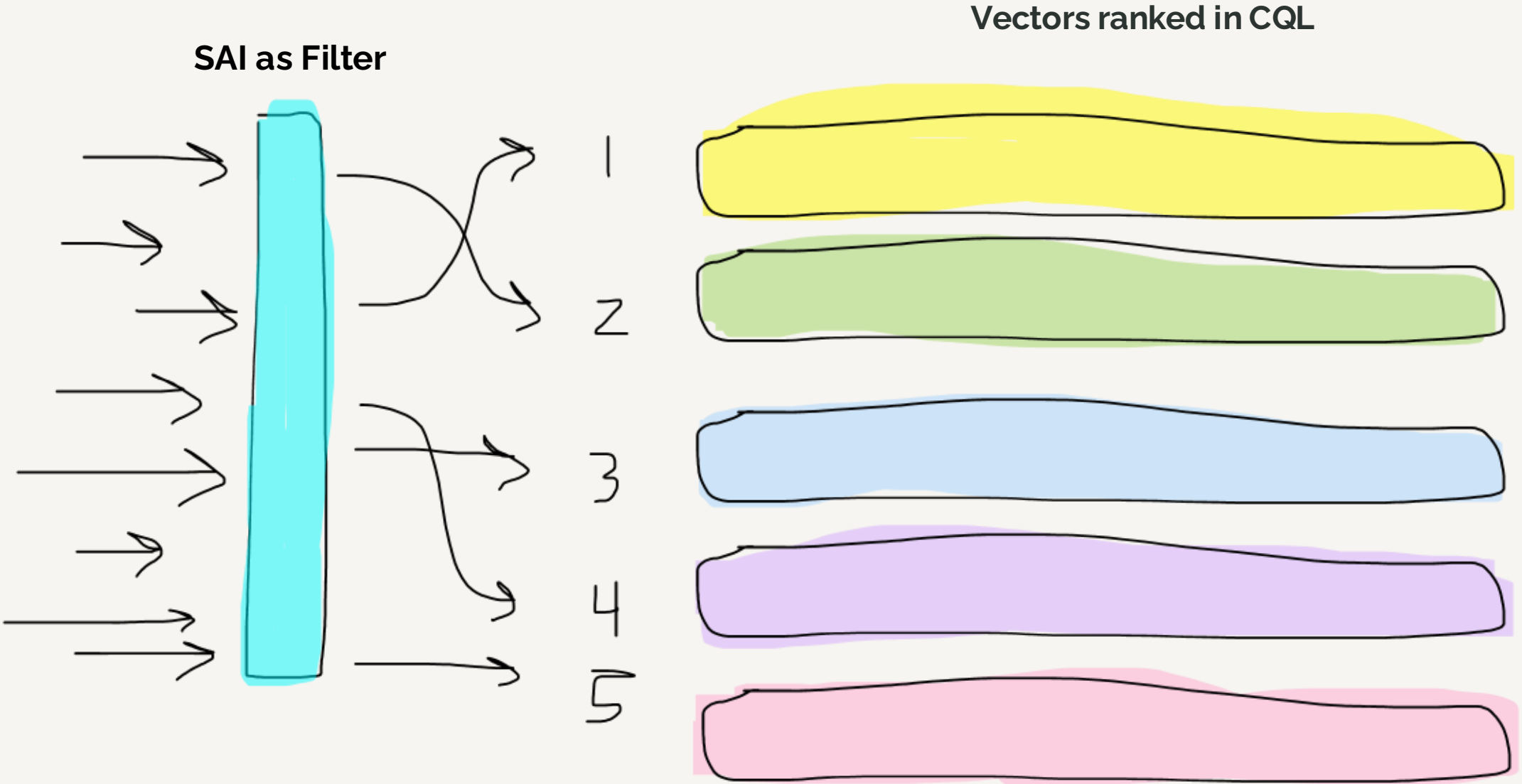
Graph RAG



Vector RAG



Hybrid retrieval & Cassandra



Setup & Queries

```
CREATE TABLE IF NOT EXISTS
hybrid_retrieval.docs (
  id text PRIMARY KEY,
  title text,
  body text,
  category text,
  keywords set<text>,
  model text,
  model_year int,
  embedding vector<float, 384>
);
```

- Similar setup to the vector example
- Now we add indexes to useful information that we'll use to filter results

Indexing on the columns

```
CREATE INDEX docs_embedding_sai ON docs (embedding) USING 'sai'  
WITH OPTIONS = {'similarity_function': 'COSINE'};
```

```
CREATE INDEX docs_keywords_sai ON docs (keywords) USING 'sai';
```

```
CREATE INDEX docs_category_sai ON docs (category) USING 'sai'  
WITH OPTIONS = {'case_sensitive': 'false', 'normalize': 'true'};
```

```
CREATE INDEX docs_model_sai ON docs (model) USING 'sai';
```

Let's ask these questions again...

Hybrid Retrieval

"Is there a recall on the tailgate?"

```
(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % python src/demo.py "What is recall 24V-330?"
```

Still get the same answer with hybrid searching

Hybrid Retrieval

"Is there a recall on the tailgate?"

QUERY: Is there a recall on the tailgate?

Exact tokens: recall, tailgate

Selective token: (none) Inferred category: recalls

--- 1) Hybrid: WHERE category = 'recalls' + ANN ---

1. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch score=0.7767
2. [recalls] recall-23v-011 [1500 2023] Recall 23V-011 instrument cluster score=0.7032
3. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera score=0.6935
4. [recalls] recall-23v-145 [2500 2023] Recall 23V-145 brake booster score=0.6934
5. [recalls] recall-22v-078 [2500 2022] Recall 22V-078 trailer brake module score=0.6808

--- 2) Keyword grounding: keywords CONTAINS ---

1. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch matched 2 tokens
2. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera matched 1 token
3. [recalls] recall-22v-311 [1500 2022] Recall 22V-311 headlight wiring harness matched 1 token
4. [recalls] recall-24v-207 [1500 2024] Recall 24V-207 fuel pump relay matched 1 token
5. [recalls] recall-24v-402 [1500 2024] Recall 24V-402 steering column bolt matched 1 token

--- 3) Vector ANN: ORDER BY embedding ANN OF ---

1. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch score=0.7767
2. [recalls] recall-23v-011 [1500 2023] Recall 23V-011 instrument cluster score=0.7032
3. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera score=0.6935
4. [recalls] recall-23v-145 [2500 2023] Recall 23V-145 brake booster score=0.6934
5. [recalls] recall-22v-078 [2500 2022] Recall 22V-078 trailer brake module score=0.6808

Still get the same answer with hybrid searching

Now, an exact recall ID grounds our answer

What is recall 24V-330?

```
(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % python src/demo.py "What is recall 24V-330?"  
□
```

Previously we didn't get this answer right with vector search alone

Now, an exact recall ID grounds our answer

What is recall 24V-330?

QUERY: What is recall 24V-330?

Exact tokens: recall, 24v-330

Selective token: 24v-330 Inferred category: recalls

--- 1) Hybrid: WHERE keywords CONTAINS '24v-330' AND category = 'recalls' + ANN ---

1. [recalls] recall-24v-330 [1500 2024] Recall 24V-330 rear axle seal score=0.7332

--- 2) Keyword grounding: keywords CONTAINS ---

1. [recalls] recall-24v-330 [1500 2024] Recall 24V-330 rear axle seal matched 2 tokens

2. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera matched 1 token

3. [recalls] recall-22v-311 [1500 2022] Recall 22V-311 headlight wiring harness matched 1 token

4. [recalls] recall-24v-207 [1500 2024] Recall 24V-207 fuel pump relay matched 1 token

5. [recalls] recall-24v-402 [1500 2024] Recall 24V-402 steering column bolt matched 1 token

--- 3) Vector ANN: ORDER BY embedding ANN OF ---

1. [recalls] recall-23v-011 [1500 2023] Recall 23V-011 instrument cluster score=0.7871

2. [recalls] recall-24v-207 [1500 2024] Recall 24V-207 fuel pump relay score=0.7701

3. [recalls] recall-23v-088 [1500 2023] Recall 23V-088 backup camera score=0.7623

4. [recalls] recall-24v-113 [1500 2024] Recall 24V-113 tailgate latch score=0.7560

5. [recalls] recall-24v-155 [2500 2024] Recall 24V-155 glow plug controller score=0.7514

Previously we didn't get this answer right with vector search alone

Vague questions = uncertain answers

How much can the Summit 1500 tow?

```
(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % python src/demo.py "How much can the Summit 1500 tow?"
```

Vague questions = uncertain answers

How much can the Summit 1500 tow?

QUERY: How much can the Summit 1500 tow?
Exact tokens: summit, 1500, tow
Selective token: (none) Inferred category: towing

--- 1) Hybrid: WHERE category = 'towing' + ANN ---

1. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year score=0.8990
2. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year score=0.8853
3. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity score=0.8791
4. [towing] hitch-classes Hitch receiver classes and weight distribution score=0.7550
5. [towing] trailer-brake-controller When a trailer brake controller is required score=0.7282

--- 2) Keyword grounding: keywords CONTAINS ---

1. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year matched 3 tokens
2. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year matched 3 tokens
3. [specs] summit-1500-2024-payload [1500 2024] 2024 Summit 1500 payload capacity matched 2 tokens
4. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity matched 2 tokens
5. [towing] trailer-brake-controller When a trailer brake controller is required matched 1 token

--- 3) Vector ANN: ORDER BY embedding ANN OF ---

1. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year score=0.8990
2. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year score=0.8853
3. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity score=0.8791
4. [specs] summit-1500-2024-payload [1500 2024] 2024 Summit 1500 payload capacity score=0.8276
5. [towing] hitch-classes Hitch receiver classes and weight distribution score=0.7550

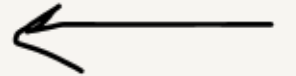
Let's try it again

```
(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % python src/demo.py --model 1500 --model-year 2024 "How much can the Summit 1500 tow?"
```

Let's try it again

```
[(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % python src/demo.py --model 1500 --model-year  
2024 "How much can the Summit 1500 tow?"  
Loading weights: 100%|██████████████████████████████████████████████████████████████| 103/103 [00:00<00:00, 10023.28it/s]  
  
=====
```

```
QUERY: How much can the Summit 1500 tow?  
Exact tokens: summit, 1500, tow  
Selective token: (none)   Inferred category: towing  
  
--- 1) Hybrid: WHERE category = 'towing' AND model = '1500' AND model_year = 2024 + ANN ---  
1. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year score=0.8990  
  
--- 2) Keyword grounding: keywords CONTAINS ---  
1. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year matched 3 tokens  
2. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year matched 3 tokens  
3. [specs] summit-1500-2024-payload [1500 2024] 2024 Summit 1500 payload capacity matched 2 tokens  
4. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity matched 2 tokens  
5. [towing] trailer-brake-controller When a trailer brake controller is required matched 1 token  
  
--- 3) Vector ANN: ORDER BY embedding ANN OF ---  
1. [towing] summit-1500-2024-tow [1500 2024] Tow ratings by configuration, 2024 model year score=0.8990  
2. [towing] summit-1500-2023-tow [1500 2023] Tow ratings by configuration, 2023 model year score=0.8853  
3. [towing] summit-2500-2024-tow [2500 2024] 2024 Summit 2500 heavy duty towing capacity score=0.8791  
4. [specs] summit-1500-2024-payload [1500 2024] 2024 Summit 1500 payload capacity score=0.8276  
5. [towing] hitch-classes Hitch receiver classes and weight distribution score=0.7550  
(.venv) mm11621@mm11621-mac-0 cassandra-hybrid-retrieval-demo % |
```



Result

```
{
  "id": "summit-1500-2024-tow",
  "title": "Tow ratings by configuration, 2024 model year",
  "category": "towing",
  "model": "1500",
  "model_year": 2024,
  "keywords": ["summit", "1500", "2024", "tow", "towing", "max tow"],
  "body": "Ratings for the 2024 Summit 1500 vary by driveline. With the 3.5L V6, 3.73 axle, and Max Tow Package, the published maximum is 11,300 lb against a GCWR of 17,000 lb. Lower axle ratios and the base cooling package derate this figure."
},
```

Takeaways

*identifiers
need filters*

*descriptions
need vectors*

*underspecified
questions
need context*

Pros

- Reduced architectural complexity. One round trip to query one table with SAI + ANN
- Adding more indexes is relatively lightweight
- Data can be embedded at write time and alongside its other column data.

Cons

- Can't do multi-hop traversal: if you can't express the path as a fixed set of SAI predicates, every hop in Cassandra becomes another round trip plus an app-side join.
- Can't do BM25, Vectors can't do negation, general limitation
- Filtering may not be enough: e.g. "what themes are happening each quarter" – need Graph RAG

Why use Cassandra for vector search

- **Context you already store:** e.g. color, make, or model year, etc. scopes results without a reranking pass.
- **Scale you already run:** the retrieval path inherits your existing partitioning and multi-region replication.
- **Retrieval beside operational data:** embeddings sit in the same row as the record, so there's no sync lag between a write and what RAG can see.
- **One table, one query, one system:** no separate vector store or search cluster to provision, secure, and keep consistent.
- **Fresh writes are immediately retrievable:** nothing waits on a batch reindex into a downstream service.

Use Cases

- **RAG chatbots and assistants** — Store document chunks and their embeddings in the same table
- **Semantic search in an existing app** — Let users search tickets, product catalogs, or knowledge-base articles by meaning while still restricting results with normal partition keys and SAI predicates like tenant or status.
- **Recommendations and "more like this"** — Compare a user or item embedding against the catalog to surface similar products, articles, or profiles without standing up a separate vector database.
- **Deduplication and fraud/anomaly detection** — Flag near-duplicate listings, reviews, or transaction patterns that differ in wording or details but sit close together in embedding space.
- **Multimodal retrieval** — Index image, audio, or video embeddings the same way as text, so image-to-image or cross-modal lookups run through the same SAI ANN path.

Cassandra 6

- ***alpha1*** March 2026, ***alpha2*** August 2026. 6.0 is not yet GA
- The work since 5.0 on vector search and SAI has been focused on correctness and latency
 - CASSANDRA-21062 – Vector Search Modernization
- Cassandra 6 brings many other changes that help make the database easier to manage and operate: ACID transactions, Constraints, Transactional Cluster Metadata, Automated Repair, and more

How do you rate your experience?



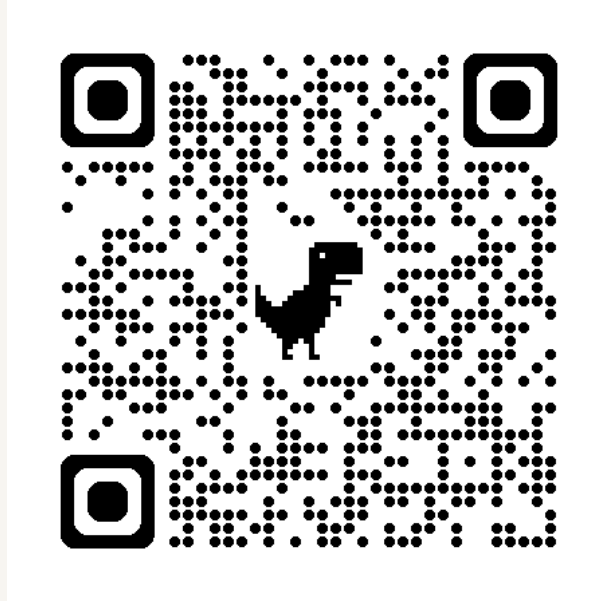
Thanks!

SLIDES: <https://github.com/mamclaughlin/2026-nyc-open-data-user-group>

VECTOR & HYBRID APPS: <https://github.com/instacluster/code-samples/tree/main/cassandra>



Slides



Demo



Connect with me!

Slides: <https://github.com/mamclaughlin/2026-nyc-open-data-user-group>

Vector & Hybrid Apps: <https://github.com/instacluster/code-samples/tree/main/cassandra>